Name: ____________________________ Partner: _______________________ Period: _________

Tier R — Remediate: Using a Table to Evaluate Inverse Trig

Use the reference table below to answer questions 1–6.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
$-\frac{\pi}{2}$	-1	0	undefined
$-\frac{\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
$-\frac{\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
$-\frac{\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	undefined
$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
π	0	-1	0

Remember: arcsin outputs an angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$; arccos in $[0, \pi]$; arctan in $(-\frac{\pi}{2}, \frac{\pi}{2})$.

1. $\arcsin\left(\frac{\sqrt{3}}{2}\right) =$ ___________ **2.** $\arccos\left(\frac{1}{2}\right) =$ ___________ **3.** $\arctan(\sqrt{3}) =$ ___________

4. $\arcsin(-1) =$ ___________ **5.** $\arccos(-1) =$ ___________ **6.** $\arctan(0) =$ ___________

Tier A — Apply: Domain, Range, and Compositions

7. Complete the table.

Function	**Domain**	**Range**
$f(x) = \arcsin x$	________________	________________
$g(x) = \arccos x$	________________	________________
$h(x) = \arctan x$	________________	________________

8. Evaluate $\sin\!\left(\arccos \frac{3}{5}\right)$.
Hint: Let $\theta = \arccos \frac{3}{5}$. Then $\cos\theta = \frac{3}{5}$. Draw a right triangle and label the sides.

9. Evaluate $\tan\!\left(\arcsin \frac{5}{13}\right)$.

Tier E — Extend: Compositions and Domain Analysis

10. Evaluate each expression. Show all work.

(a) $\sin\!\left(\arcsin \frac{1}{2}\right) =$ ______________

(b) $\arcsin\!\left(\sin \frac{\pi}{3}\right) =$ ______________

(c) $\arcsin\!\left(\sin \frac{2\pi}{3}\right) =$ ______________

(d) $\cos\!\left(\arccos(-0.7)\right) =$ ______________

11. For which values of x does $\sin(\arcsin x) = x$? Explain.

12. For which values of x does $\arcsin(\sin x) = x$? Explain why the answer is not "all real x."

13. Simplify $\cos(\arctan x)$ for $x > 0$. Express your answer without any trig or inverse trig functions. (*Hint:* Let $\theta = \arctan x$; use a right-triangle diagram.)